

VISHESH RAJU

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Software Engineer Experienced in LLM orchestration, RAG pipelines, and prompt engineering for AI-powered applications, AI-driven solutions, NLP, and Generative AI applications, and scalable AWS architecture, with hands-on experience in microservices architecture, big data processing, and delivering measurable operational excellence.

EDUCATION

New Jersey Institute of Technology

Newark, NJ

Master of Science in Computer Science; **GPA: 3.9/4.0**

Aug. 2024 – May 2026

- Coursework: Data Structures & Algorithms, Big Data, Machine Learning, Deep Learning, Operating Systems, Database System Design, Web Development, Data Analytics, Data Mining, Software Engineering

R N Shetty Institute of Technology

Bengaluru, India

Bachelor of Engineering in Artificial Intelligence and Machine Learning

Dec. 2020 – May 2024

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL

Frameworks & Libraries: React, Node.js, Express, FastAPI, Flask, PyTorch, TensorFlow, Pandas, NumPy, OpenCV

Cloud & Infrastructure: AWS (EC2, S3, Lambda, BedRock, API), Docker, Kubernetes, CI/CD

Big Data & Databases: Hadoop, MapReduce, Apache Oozie, MongoDB, PostgreSQL, MySQL, NoSQL, SQLite

Tools & Practices: Git, GitHub Actions, REST API, Distributed Systems, Agile, System Design, OOP, LLMs, RAG, Prompt Engineering

EXPERIENCE

Research Data Analyst

May 2025 – Present

New Jersey Institute of Technology - Center for Injury Biomechanics (CIBM3)

Newark, NJ

- Built Python microservices to automate 3D microscopy data processing, cutting manual effort by **80% and achieving 2.3x throughput on 200GB+ datasets**
- Refactored legacy analysis pipeline** into modular production-ready components, **improving reliability by 15%**; set up CI/CD with automated testing and reproducible environments
- Developed data mining pipelines to extract spatial and temporal patterns from large-scale unstructured scientific datasets

Software Engineering Intern

Oct. 2023 – Dec. 2023

22by7 Technologies Ltd.

Bengaluru, India

- Built a predictive maintenance ML model achieving **97% accuracy**, reducing unplanned equipment downtime in production
- Owned end-to-end ML pipeline data preprocessing, feature engineering, model training, and production integration

PROJECTS

VetGraph — Knowledge Graph Library | Python, LLMs, NetworkX, OpenAI API

- Built a provider-agnostic Python library that transforms unstructured text into queryable Knowledge Graphs using LLM orchestration, prompt engineering, and NetworkX-based graph analysis
- Gained 200+ downloads in the first week of release; supports multiple LLM backends.

AuRA Discovery — AI Research Assistant | Python, LLMs, FastAPI

- Developed a multi-modal Drug–Drug Interaction (DDI) prediction system integrating LLMs, Neo4j graphs, and hybrid ML/GNN models, achieving 0.86 F1-score and 0.88 PR-AUC on biomedical datasets.
- Engineered scalable feature pipelines using Python, PyTorch, XGBoost, Hadoop, and Neo4j to process 5,000+ biomedical interaction records and generate 5,900+ predictive features.

FlowBoard | React, TypeScript, Spring Boot, PostgreSQL, AWS, API

- Developed a full-stack AI-assisted Kanban and project collaboration platform with React, Spring Boot, PostgreSQL, JWT authentication, context optimization, and real-time WebSocket updates
- Deployed the frontend on AWS Amplify and backend APIs on AWS Lambda with API Gateway integration, using AWS S3 for storage and GitHub Actions for CI/CD automation

Distributed Crime Analytics Pipeline | Java, Hadoop, MapReduce, Apache Oozie, AWS EC2

- Processed 8.2M crime records on a 12-node AWS EC2 Hadoop cluster using MapReduce to extract crime trends, temporal patterns, and geographic hotspots at scale
- Orchestrated the full pipeline with Apache Oozie for automated job scheduling and dependency management